```python
import numpy as np
import pandas as ppd
import matplotlib.pyplot as plt

dataset = ppd.read_csv(
    '/content/Ads_CTR_Optimisation.csv'
)

print(dataset.describe())

import random

N= 10000
d = 10

numbers_of_selections = [0] * d
sums_of_rewards = [0] * d
ads_selected = []
total_reward = 0

epsilon = 0.1

mean_rewards = dataset.describe().loc['mean'].values

for n in range(0, N):
  ad = 0

  if random.uniform(0, 1) < epsilon:
    ad = random.randrange(d)

  else:
    ad = np.argmax(mean_rewards)

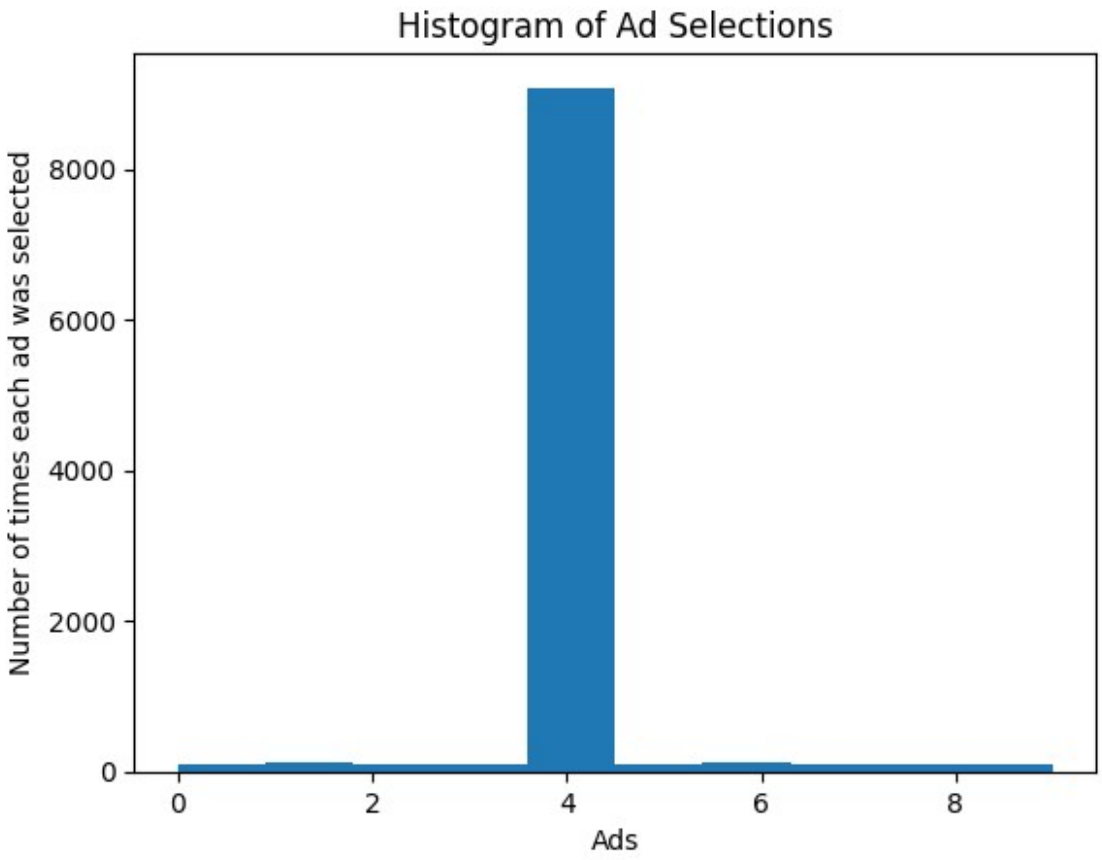
  ads_selected.append(ad)
  reward = dataset.values[n, ad] # Get the reward for the selected ad
  numbers_of_selections[ad] += 1
  sums_of_rewards[ad] += reward
  total_reward += reward

# Visualization of the results
plt.hist(ads_selected)
plt.title('Histogram of Ad Selections')
plt.xlabel('Ads')
plt.ylabel('Number of times each ad was selected')
plt.show()
```

```
                Ad 0          Ad 1          Ad 2          Ad 3
Ad 4  \
count  10000.000000  10000.000000  10000.000000  10000.000000
10000.000000
mean       0.170300      0.129500      0.072800      0.119600
```

```
0.269500
std        0.375915      0.335769      0.259821      0.324509
0.443722
min        0.000000      0.000000      0.000000      0.000000
0.000000
25%        0.000000      0.000000      0.000000      0.000000
0.000000
50%        0.000000      0.000000      0.000000      0.000000
0.000000
75%        0.000000      0.000000      0.000000      0.000000
1.000000
max        1.000000      1.000000      1.000000      1.000000
1.000000

                Ad 5          Ad 6          Ad 7          Ad 8
Ad 9
count  10000.000000  10000.000000  10000.000000  10000.000000
10000.00000
mean       0.012600      0.111200      0.209100      0.095200
0.04890
std        0.111546      0.314395      0.406686      0.293506
0.21567
min        0.000000      0.000000      0.000000      0.000000
0.00000
25%        0.000000      0.000000      0.000000      0.000000
0.00000
50%        0.000000      0.000000      0.000000      0.000000
0.00000
75%        0.000000      0.000000      0.000000      0.000000
0.00000
max        1.000000      1.000000      1.000000      1.000000
1.00000
```


Histogram of Ad Selections
Number of times each ad was selected
8000
6000
4000
2000
0
0
2
4
6
8
Ads